

Bell's Theorem Revisited: Structural Collapse from Entropy Geometry

David Sigtermans

August 2025

Abstract

Bell's theorem is widely cited as definitive evidence that the physical world is forced to choose: either locality must be abandoned, or realism must yield. Yet this dilemma conceals a deeper structural assumption—one rarely made explicit. The standard argument presupposes a *flat geometry of resolution*: it assumes that the space of possibilities is already perfectly divided, every alternative sharply distinguishable, and all correlations defined on an informationally uniform background.

The Total Entropic Quantity (TEQ) framework overturns this unexamined starting point. In TEQ, locality, realism, and even the act of measurement itself arise from the curved geometry of entropy-weighted resolution, not as metaphysical postulates but as structural consequences. Distinguishability is not given, but selected: it is a product of entropy geometry, and can be sharply limited or curved.

This paper revisits Bell's inequality in this new light. Collapse is reframed—not as a discontinuous event or an observer-induced mystery, but as the stabilization of resolution under entropy curvature. We show that the statistical structure of Bell-type experiments is not a paradox, but the signature of curved entropy geometry selecting resolution-stable paths. No hidden variables, no special role for the observer, and no violation of empirical predictions are required. Instead, what appears as "nonlocality" is reinterpreted as evidence for shared entropy curvature; what appears as a quantum mystery becomes a geometric fact.

By deriving Bell-type violations from first principles within TEQ, we preserve all experimental results while providing a unified, falsifiable, and conceptually grounded explanation. The flatness assumption at the heart of the standard reading is not denied—it is generalized, allowing the theory to recover known physics as a limit, while predicting possible deviations in regimes of extreme entropy curvature. This approach reframes one of the deepest puzzles of quantum theory as a question of geometry, not ontology.

Keywords: entropy geometry; Bell's theorem; quantum nonlocality; resolution metric; entropy-weighted action; TEQ framework; structural collapse; entropy curvature; measurement problem; factorization breakdown; curvature-induced correlations; quantum foundations; minimal principle.

Meta-Abstract:

This paper provides a structural reformulation of Bell’s theorem from within the Total Entropic Quantity (TEQ) framework. The argument proceeds not by reinterpretation, but by systematic derivation: all key results—including the breakdown of Bell’s factorization condition, the emergence of quantum correlations, and the resolution of the measurement problem—are derived from a geometric principle in which entropy curvature governs distinguishability. The paper makes the following explicit:

- **Foundational principles:** TEQ begins from two core axioms—entropy geometry as a generative constraint, and a minimal principle selecting stable distinctions—explicitly defined and referenced in Sections 3 and [6, 1].
- **Logical pathway:** All classical and quantum measurement results, including Bell-type violations, are shown to arise from the entropy-weighted action (Eq. (3)), without assuming projection, wavefunction collapse, or hidden variables.
- **Derivational structure:** The structure of resolution fixpoints, the breakdown of probabilistic factorization, and the suppression of unstable alternatives are justified via the entropy curvature of the resolution metric. Worked examples and explicit models (Sections 3.3–3.4) illustrate the path from first principles to testable predictions.
- **Empirical and interpretive scope:** All standard quantum predictions are recovered as a limiting case (flat entropy geometry); possible deviations are identified for regimes with strong entropy curvature, providing avenues for falsifiability (see Section 3.4 and Section 4).
- **Comparative context:** Connections and contrasts with historical approaches (von Neumann, Wigner, Grete Hermann) and modern interpretations (Bohmian, GRW, QBism) are explicitly drawn, clarifying how TEQ subsumes, reduces, or replaces standard postulates.
- **Philosophical synthesis:** A closing postlude distills the structural implications of the derivation, showing how TEQ reframes collapse, realism, and nonlocality as consequences of entropy geometry rather than metaphysical assumptions.

The manuscript is structured for clarity: each section begins with an intuitive overview, key parameters are explained in context, and derivational steps are cross-referenced throughout. This meta-abstract serves to orient the reader—expert or non-expert—to the deductive structure, scope, and empirical significance of the TEQ approach to Bell’s theorem.

Term / Notation	Definition / Meaning
TEQ	<i>Total Entropic Quantity</i> framework; physical theory in which structure, measurement, and correlation arise from entropy-weighted resolution geometry, rather than as primitive assumptions.
$S_{\text{eff}}[\phi]$	Entropy-weighted effective action: $S_{\text{eff}}[\phi] = \int dt \left(L(\phi, \dot{\phi}) - i\hbar\beta g(\phi, \dot{\phi}) \right)$
	selects trajectories stable under entropy flow; central variational principle in TEQ.
$L(\phi, \dot{\phi})$	Standard Lagrangian; encodes kinetic and potential energy in classical or quantum dynamics.
$g(\phi, \dot{\phi})$	Resolution metric or entropy flux functional; quantifies the local entropic cost of maintaining distinguishability along a path. Derived from the geometry of entropy.
β	Entropy curvature sensitivity parameter; modulates the weighting of entropy cost in S_{eff} , acting as a selector for resolution-stable paths.
δg_{shared}	Shared entropy curvature term; captures coupling between subsystems in the resolution metric, leading to correlated probabilities and breakdown of factorization.
Resolution fixpoint	A configuration or outcome stabilized by entropy-weighted dynamics; a “collapse” not as a physical event but as an entropic selection principle.
Factorization breakdown	Failure of the standard product rule for joint probabilities (e.g., Bell’s condition); signals presence of shared entropy curvature in configuration space.
Curvature-induced correlations	Nonlocal-looking statistical correlations arising not from signals, but from shared entropy geometry; central mechanism for Bell-type violations in TEQ.
Measurement problem (in TEQ)	The classical puzzle of quantum “collapse” reframed as selection of resolution fixpoints by entropy geometry, without reference to observers or projection.
Flat entropy geometry	Special case where the resolution metric is uniform; recovers standard quantum statistics and classical locality as limiting behaviors.
Minimal principle	Core axiom in TEQ: structure is selected so as to maximize stability under entropy flow, subject to geometric constraints.
Hidden variable (λ)	In standard Bell arguments, a hypothetical parameter encoding all “complete” information; in TEQ, its presumed flat structure is replaced by a curved entropy geometry.
Path integral (in TEQ)	The sum over all distinguishable, entropy-weighted trajectories; generalizes quantum path integrals by incorporating entropy curvature and resolution constraints.

Table 1: Glossary of core terms and notations (TEQ/Bell context).

1 Introduction: Bell’s Theorem and the Geometry of Collapse

Bell’s theorem is one of the most influential results in quantum foundations. It appears to force a fundamental choice between two deeply held intuitions: *locality*, the idea that physical influences cannot travel faster than light, and *realism*, the view that physical properties exist independently of measurement. Countless experimental confirmations of Bell inequality violations have led many physicists to conclude that one of these must be false. Either the world is nonlocal, or reality is not fully real until observed.

But beneath this dramatic framing lies a subtler, more foundational assumption—one that is rarely questioned. Bell’s argument, and the standard interpretation of its violation, rest on an implicit belief in a kind of **informational flatness**: that the space of all possible outcomes, hidden variables, and correlations is globally and uniformly distinguishable. This means that every possible alternative is as crisply separated from every other as squares on a perfect chessboard, or pixels in a high-resolution image—the underlying “canvas” of possibilities is assumed to be flat and unwrinkled, with every configuration perfectly divided from the next.

In physical terms, *flatness* here refers to a kind of perfect resolution: no outcome is ambiguous or blurred; any distinction that can be drawn, is already present and cost-free. The configuration space—the mathematical setting for all possible states of a system—is imagined as a uniform, featureless landscape, where drawing sharp boundaries between alternatives is trivial.

The **Total Entropic Quantity (TEQ)** framework challenges this hidden assumption. It begins from a different principle: what can be distinguished—what we resolve as separate, independent, or correlated—is not fixed, but dynamically selected. In TEQ, *distinguishability* is an emergent property of **entropy geometry**: a structural field that governs which alternatives remain resolvable as entropy evolves across configuration space. *Here, “flow of entropy” refers to how entropy—understood as a measure of unresolved or inaccessible information—varies along system trajectories. Rapid entropy increase suppresses fine distinctions; slow or stationary entropy permits sharper resolution. The geometry is “curved” because this resolvability varies across space and direction, constrained by local entropy gradients.*

In TEQ, physical structure and the possibility of measurement are not assumed from the outset. They emerge from the geometry of resolution itself—specifically, from how entropy gradients constrain which distinctions can persist. Locality, realism, and measurement are not added as interpretive choices; they arise as structural consequences of the system’s entropy dynamics [1].

This perspective opens a third path in the debate about Bell’s theorem. Rather than choosing between hidden variables and observer-induced collapse, we suggest that both views miss the underlying structure. The violation of Bell inequalities does not demand nonlocality or subjectivity. Instead, it signals that the assumption of globally flat resolution is too strong. What looks like “instantaneous influence” or “wavefunction collapse” is the geometric effect of shared entropy curvature—a kind of “wrinkle” or coupling in the resolution structure that makes certain correlations structurally unavoidable, even when no signals pass between

distant systems.

In this paper, we rederive the statistical structure of Bell-type experiments from within the TEQ framework. Here, “collapse” is not a discontinuous event, nor an epistemic update, nor a quantum jump. It is the stabilization of a **resolution fixpoint** under entropy constraints—a kind of attractor in the entropy-weighted landscape, where making further distinctions becomes entropically prohibitive. All empirical predictions of quantum mechanics are preserved, but their interpretation is restructured.

Our goal is not to reinterpret quantum theory from a different philosophical angle, but to *rederive its structural features from a deeper geometric principle*. In doing so, we expose the hidden flatness assumption that underlies standard treatments of Bell’s theorem, and provide an alternative framework in which quantum correlations are the expected result of entropy-driven geometric filtering—not violations of locality, nor enigmas of consciousness.

Roadmap: The sections that follow will make this argument precise. We begin by dissecting the hidden assumptions in the derivation of Bell’s inequality. We then introduce the entropy geometry that underlies TEQ, and show how resolution curvature replaces the need for nonlocal signaling. Finally, we analyze the consequences for quantum statistics, falsifiability, and future experimental tests, aiming to make each technical step accessible both in mathematical form and physical intuition.

2 What Bell’s Inequality Assumes

Bell’s inequality is often presented as a purely mathematical result, derived from seemingly minimal assumptions about locality and realism. The standard logic is crisp: if the outcomes of measurements on spatially separated systems depend only on local settings and shared hidden variables, then certain inequalities must hold. The experimental violations of those inequalities are then taken to force a radical revision of our worldview.

But this presentation masks a deeper layer of assumptions—assumptions about the structure of information, the topology of configuration space, and the geometry of distinguishability. Beneath the algebra lies a specific conception of what it means for variables to be *independent*, what it means for outcomes to be *defined*, and what it means for a theory to *account for correlations*. At the heart of Bell’s derivation is an assumption of **flat resolution**: that all variables live in a globally well-defined space of distinctions, where every alternative is crisply separable from every other.

Sidebar: What is Informational Flatness?

“Informational flatness” means that every outcome, alternative, or state is as sharply defined and accessible as any other. It is as if the entire space of possibilities were a perfectly woven grid, like a bolt of fine cloth where every thread is separate and untangled. By contrast, in a “curved” informational landscape, distinctions blur or dissolve. Some outcomes become difficult or impossible to separate—like details fading in a low-resolution or overexposed image, or signals merging in a radio filled with static. The structure of resolution is no longer flat and uniform, but shaped by local constraints that limit what can be clearly distinguished.

This assumption is not merely mathematical. It encodes a worldview in which the physical state of the universe at any moment—represented by a hidden variable λ —can be described as a point in a fully resolvable configuration space. In such a space, distinctions between different possible worlds are absolute, globally defined, and simultaneously available.

To see this explicitly, consider the standard setup of a Bell-type experiment. Two particles are emitted from a common source and sent to spatially separated measurement stations. At each station, an experimenter chooses a measurement setting: a for particle A, and b for particle B. The resulting outcomes are denoted $A \in \{+1, -1\}$ and $B \in \{+1, -1\}$. A shared hidden variable λ is introduced to encode any underlying properties or common causes that might determine these outcomes. It is drawn from a probability distribution $\rho(\lambda)$.

The core assumption of local realism is then expressed as the factorizability condition:

$$P(A, B|a, b, \lambda) = P(A|a, \lambda) P(B|b, \lambda). \quad (1)$$

This relation concerns conditional probabilities: it asserts that once the shared variable λ is known, the joint probability of outcomes A and B factorizes into independent local terms. That is, A depends only on the local setting a , and B only on b , with no influence or information flow between the two wings of the experiment.

But more deeply, this assumption encodes a particular geometric stance: that the resolution structure of the system is flat. The hidden variable λ is presumed to live in a space where all distinctions—between outcomes, settings, and subsystems—are already well-separated and informationally clean. The factorizability condition is therefore not merely a constraint on causal influences, but a structural assumption about the geometry of distinguishability itself: that the configuration space admits global separation of alternatives without entropic cost.

Historical and Conceptual Background

The idea that such a flat space underlies physical measurement was implicit in the early formalizations of quantum theory. Grete Hermann argued in the 1930s that causal relations in quantum mechanics depend on the algebraic structure in which events are embedded—that causality itself is not a primitive, but a relational, structurally constrained notion [2, 3]. Von Neumann’s Hilbert space formalism gave this relational structure a precise mathematical home, but at the cost of embedding quantum theory within a globally defined geometric framework of pre-assumed resolution: one where all distinctions are presupposed rather than derived [4]. Wigner’s later work on symmetries made this geometric commitment more explicit, connecting transformations and conservation laws to the assumed structure of configuration space [5].

Yet, despite these developments, the assumption that distinguishability is absolute and globally valid has rarely been questioned. Even in quantum foundations, it is often treated as a given: probabilities are defined over a fixed configuration space; measurements reveal values; and the observer’s role is to select among pre-defined alternatives.

The TEQ Perspective

The **Total Entropic Quantity (TEQ)** framework challenges this hidden flatness. In TEQ, distinctions are not absolute—they are structural consequences of entropy geometry. The ability to resolve outcomes or treat configurations as independent is governed by a curved resolution metric that reflects the local entropic cost of maintaining distinctions. Where entropy curvature is steep, fine-grained separation becomes unstable. As a result, factorizability—and the derivation of Bell inequalities—holds only in the limiting case of flat entropy geometry, where the cost of distinction is uniform across configuration space.

This reinterpretation is developed rigorously in Section 3 and in [6], where the entropy-weighted action functional selects resolution-stable paths and penalizes configurations that require high entropic cost to sustain. Bell’s derivation, viewed through this lens, assumes that such costs are negligible—that the configuration space has no curvature. But this is not generically true.

2.1 Flat Geometry and Predetermined Outcomes

Let us return to Eq. (1), the key assumption behind Bell’s theorem. It claims that once the hidden variable λ is specified, the outcome at one detector is conditionally independent of the setting and outcome at the other. This expresses locality. But it also assumes something stronger: that λ itself is defined over a space of fully distinct alternatives, and that each possible measurement outcome corresponds to a unique, pre-existing property of that hidden state.

In other words, Eq. (1) presupposes that all relevant variables are drawn from an information geometry with **no resolution gradient**. The space in which λ lives is flat: every possible value of λ supports distinctions that are equally sharp, equally accessible, and equally independent. There are no “blurred” regions, no ambiguous neighborhoods, no entropic penalty for maintaining distinctions.

The TEQ framework rejects this view. It replaces the flat canvas of classical hidden-variable models with a curved geometry of entropy. In TEQ, distinguishability is not uniform across configuration space. Some distinctions are harder to sustain; some are filtered out by entropy flow. What we call “measurement” does not reveal a value that already existed, but selects a **resolution fixpoint**—a configuration that is stable under entropy-weighted dynamics (see Eq. (5)). The act of measurement is not passive revelation, but active selection within a curved space of distinguishability.

Thus, the assumption that λ contains fully separable, independent substructures already embeds the conclusion that TEQ denies: that distinguishability is globally and uniformly real. The factorizability condition in Eq. (1) is not neutral. It imports a geometric model that TEQ sets out to replace.

Experiments inspired by Bell’s theorem, including the landmark tests by Aspect et al. [7], have repeatedly demonstrated violations of the inequality:

$$|E(a, b) - E(a, b')| + |E(a', b) + E(a', b')| \leq 2. \quad (2)$$

Here, $E(a, b)$ denotes the expectation value of the product of outcomes A and B when measurement settings a and b are used. The parameters a and a' represent two distinct

measurement settings available to observer A (e.g., different spin or polarization directions), and b and b' are the corresponding settings for observer B. These settings are chosen independently and typically at random for each experimental trial. The inequality places a bound on the correlations that are possible under any theory satisfying local realism and the factorization condition of Eq. (1). It assumes that all possible outcomes for each setting exist simultaneously and are determined by a shared hidden variable λ .

The standard interpretation of this violation offers a stark dilemma: either we must give up *locality*—by allowing influences that travel faster than light—or we must abandon *realism*—by denying that physical properties exist prior to measurement. This tension has fueled decades of debate, with various interpretations of quantum mechanics proposing different resolutions.

But this dilemma presumes a fixed geometric background in which distinguishability is absolute and independent of context. In particular, it presumes that the configuration space over which λ , A , and B are defined is informationally flat—that all distinctions between possible outcomes are globally separable and equally accessible. TEQ questions this foundation.

In the TEQ framework, distinguishability is not absolute but shaped by entropy geometry. The ability to resolve a difference between outcomes is a structural property, not a background assumption. The configuration space is equipped with a resolution metric $g(\phi, \dot{\phi})$ that may be curved, meaning that some distinctions are easier to sustain than others, depending on entropy flow.

Under such curvature, the factorizability assumption in Eq. (1) breaks down: not due to any signal or hidden influence, but because the underlying resolution structure is shared and non-separable. When two subsystems A and B are entangled, they inhabit a region of configuration space where entropy gradients couple their trajectories. The TEQ path integral then assigns joint weights that cannot be written as independent products.

This structural breakdown is analyzed explicitly in Section 3.4, where we show how a shared entropy curvature term $\delta g_{\text{shared}}(\phi_A, \phi_B)$ naturally leads to correlations that violate Bell’s inequality. These correlations do not arise from faster-than-light communication, nor from any hidden determinism. They arise from a curved geometry of resolution that filters which joint configurations are entropically stable.

Thus, the violation of Eq. (2) is not a metaphysical crisis. It is a geometric signature: an indication that resolution is constrained by entropy flow, and that the structure of configuration space is curved. Within TEQ, what appears as paradox becomes prediction.

Conclusion

In summary, Bell’s inequality assumes more than it appears. It assumes a specific geometry of information—a flat space in which all distinctions are predefined and separable. The TEQ framework replaces this assumption with a dynamic, entropy-driven structure. From this perspective, the violation of Bell’s inequality is not a failure of locality or realism. It is a pointer to the limits of flat resolution models and a window into a deeper geometric foundation.

3 The TEQ Framework: Entropy Geometry and Effective Action

The Total Entropic Quantity (TEQ) framework reformulates the dynamics of physical systems not in terms of fixed trajectories or predefined outcomes, but in terms of the evolving geometry of distinguishability. In this view, what persists is not what is energetically favorable in the traditional sense, but what can remain resolved under entropy flow.

Standard formulations of mechanics—both classical and quantum—presuppose a configuration space in which states are distinguishable by default. The geometry of this space is typically treated as flat and fixed, and resolution is implicitly assumed to be cost-free. In TEQ, by contrast, resolution is a dynamical resource: maintaining fine distinctions between nearby configurations incurs an entropic cost. This cost is encoded by a resolution metric $g(\phi, \dot{\phi})$, which assigns to each path $\phi(t)$ a geometric measure of how difficult it is, under prevailing entropy conditions, to sustain its internal distinguishability over time.

This leads to a modified principle of path selection. Instead of minimizing action alone, the physically realized path minimizes a new quantity: the *entropy-weighted effective action*, introduced and derived in [6]:

$$S_{\text{eff}}[\phi] = \int dt \left(L(\phi, \dot{\phi}) - i\hbar\beta g(\phi, \dot{\phi}) \right). \quad (3)$$

Here:

- $L(\phi, \dot{\phi})$ is the conventional Lagrangian, encoding energetic and kinematic structure,
- $g(\phi, \dot{\phi})$ quantifies the entropic cost of maintaining resolution along the path $\phi(t)$,
- β is a global or local sensitivity parameter that amplifies entropy curvature effects (cf. Section 3.4),
- and the factor $i\hbar$ connects the formulation to quantum dynamics via complex weighting in the path integral.

This expression transforms the principle of least action into one of *least entropic instability*. The path that survives is not just one that balances mechanical constraints, but one that avoids regions of high resolution cost. This provides a structural reason for the suppression of certain alternatives—not because they are dynamically forbidden, but because they cannot maintain distinguishability under entropy flow. This principle governs collapse (Section 3.2), interference (cf. [9]), and quantum statistics (Section 4).

3.1 Resolution Geometry and Entropy Flow

In classical mechanics, the evolution of a system is governed by the variational principle: the system follows a trajectory $\phi(t)$ that extremizes the classical action $S[\phi] = \int L(\phi, \dot{\phi}) dt$. This trajectory minimizes (or extremizes) energy-based costs.

TEQ generalizes this principle. The realized path is selected not only for its dynamical properties, but for its entropic viability. Paths that require the system to sustain delicate

internal distinctions across regions of high entropy gradient are suppressed. That is, if a path passes through regions of configuration space where entropy flow rapidly erodes distinguishability, that path becomes statistically improbable.

This filtering is encoded in the resolution metric $g(\phi, \dot{\phi})$, which measures how “costly” it is—entropically and geometrically—to sustain high resolution at each point in space and time. Regions with steep entropy gradients correspond to high values of g , and thus low statistical weight in the path integral.

The probability of a path is not uniform, nor merely energy-dependent. It is shaped by the geometry of resolution:

$$\mathbb{P}[\phi] \propto \exp\left(-\beta g(\phi, \dot{\phi})\right). \quad (4)$$

This exponential suppression parallels the role of Boltzmann weights in statistical mechanics, but applies to path space rather than configuration microstates. The parameter β acts as a geometric inverse temperature: the higher its value, the more entropy curvature dominates the filtering of paths (cf. Eq. (5) and discussions in [6, 1]).

The result is not that the system “chooses” a trajectory in any agential sense. Rather, only certain paths remain dynamically visible: those that can be stably distinguished under entropy flow. Others are filtered out. Just as a river avoids climbing steep hills—not by deciding to, but because flow under gravity cannot sustain such paths—so too resolution flow avoids entropic ridges. What persists are the valleys: the entropically stable paths that survive under filtering.

This principle underlies the TEQ reinterpretation of collapse (Section 3.2), and explains how geometric structure replaces observer-driven discontinuities in Bell-type experiments (Section 3.4).

3.2 Collapse as a Resolution Fixpoint

Collapse, within TEQ, is not a discrete event imposed from outside the dynamics. It is an internal structural convergence. A configuration ϕ_* is said to be realized when it satisfies:

$$S_{\text{eff}}[\phi_*] \leq S_{\text{eff}}[\psi] \quad \text{for all admissible } \psi. \quad (5)$$

That is, it is a local minimum—or at least a stationary point—of the entropy-weighted effective action. Such configurations are **resolution fixpoints**: they are surrounded by neighboring paths that are entropically too costly to sustain. No observer is needed, and no stochastic mechanism is imposed. Collapse is simply the geometric outcome of selective pressure under entropy flow, as detailed in [1].

3.3 Entropy Curvature and the Breakdown of Factorization

Bell’s factorization condition (Eq. (1)) assumes that, once a shared hidden variable λ is known, the outcomes of measurements on subsystems A and B are independent:

$$P(A, B|a, b, \lambda) = P(A|a, \lambda) P(B|b, \lambda).$$

This implies that all relevant causal influences are contained in λ , and more subtly, that the underlying resolution geometry is separable: the capacity to distinguish between outcomes on one side is unaffected by what occurs on the other.

In geometric terms, this assumption presumes a flat configuration space in which distinguishability is globally defined and uncorrelated between subsystems. But TEQ challenges this presumption: it treats resolution as a dynamic, context-sensitive resource. When two subsystems share entropy curvature—i.e., when the cost of resolving one depends structurally on the state of the other—the configuration space becomes curved and non-separable.

In such cases, the resolution metric $g(\phi, \dot{\phi})$ acquires an interaction term:

$$g(\phi, \dot{\phi}) = g(\phi_A, \dot{\phi}_A) + g(\phi_B, \dot{\phi}_B) + \delta g_{\text{shared}}(\phi_A, \phi_B). \quad (6)$$

The first two terms describe the resolution cost of subsystems A and B independently. The third term, δg_{shared} , encodes curvature-induced coupling: a measure of how the resolution of ϕ_A is geometrically entangled with that of ϕ_B .

Under this joint structure, the entropy-weighted path amplitude—see Eq. (4)—becomes:

$$\mathbb{P}[\phi] \propto \exp \left(-\beta g(\phi_A, \dot{\phi}_A) - \beta g(\phi_B, \dot{\phi}_B) - \beta \delta g_{\text{shared}}(\phi_A, \phi_B) \right).$$

This expression clearly violates the factorization condition: it cannot be written as a product of independent terms for A and B . The resolution geometry has curvature that structurally couples the two subsystems, even in the absence of any signal or dynamical interaction.

This breakdown is not an ad hoc effect. It follows directly from the entropy-weighted action principle introduced in Section 3, and is demonstrated explicitly in the worked example of Section 3.4, where a toy model shows how shared curvature terms lead to suppressed anti-correlated outcomes. As shown there, the breakdown of Bell’s inequality (Eq. (2)) emerges as a consequence of the geometry of resolution—not a violation of relativistic causality, nor a metaphysical puzzle about hidden variables.

In this light, the violation of Bell-type inequalities is reinterpreted. It does not indicate a failure of locality or realism per se, but rather a failure of the flat-resolution assumption. As argued throughout this paper and in the foundational derivation of the entropy-weighted action [6], what survives under entropy flow are not factorized configurations, but resolution-stable paths shaped by shared entropy geometry.

3.4 Worked Example: Entropy Curvature and Bell-type Correlations

To make these ideas concrete, we present a simplified model showing how shared entropy curvature leads to Bell-type violations, as predicted in the TEQ framework. This example operationalizes the mechanism introduced in Section 3.3, using the entropy-weighted path integral formalism from Eq. (3) and the weighting rule in Eq. (4).

Two-Qubit Bell Scenario with Shared Entropy Curvature

Consider two qubits, A and B , spatially separated but entangled. Each is measured in a basis determined by settings a and b , with outcomes $A \in \{\pm 1\}$, $B \in \{\pm 1\}$. The standard

local realist model assumes:

$$P(A, B|a, b, \lambda) = P(A|a, \lambda) P(B|b, \lambda),$$

with λ encoding shared hidden information (cf. Eq. (1)).

In TEQ, by contrast, the effective action for the joint configuration $\phi = (\phi_A, \phi_B)$ includes a non-separable resolution term:

$$S_{\text{eff}}[\phi] = \int dt \left(L_A(\phi_A, \dot{\phi}_A) + L_B(\phi_B, \dot{\phi}_B) - i\hbar\beta \left[g_A(\phi_A, \dot{\phi}_A) + g_B(\phi_B, \dot{\phi}_B) + \delta g_{\text{shared}}(\phi_A, \phi_B) \right] \right), \quad (7)$$

where δg_{shared} reflects shared entropy curvature as introduced in Eq. (6).

The corresponding path amplitude becomes:

$$\mathbb{P}[\phi] \propto \exp \{ -\beta [g_A + g_B + \delta g_{\text{shared}}] \},$$

matching the entropy-weighted path integral structure of Eq. (4).

Curvature-Induced Correlations

Assume that the shared entropy curvature penalizes anti-correlated outcomes. Define:

$$\delta g_{\text{shared}}(\phi_A, \phi_B) = \gamma \delta_{A \neq B},$$

with $\gamma > 0$ quantifying the curvature-induced resolution cost. Then:

$$P_{\text{TEQ}}(A, B|a, b) \propto \int d\lambda \rho(\lambda) \exp(-\beta[g_A + g_B + \gamma \delta_{A \neq B}]) \quad (8)$$

$$= C \cdot P(A|a) P(B|b) \cdot \begin{cases} e^{-\beta\gamma} & \text{if } A \neq B, \\ 1 & \text{if } A = B, \end{cases} \quad (9)$$

where C is a normalization constant. The anti-correlated outcomes are exponentially suppressed by the curvature term. As a result, Bell's inequality (Eq. (2)) can be violated even though no nonlocal influence occurs.

Interpretation

This example illustrates how entropy geometry reshapes probability structure. In the absence of shared curvature ($\gamma = 0$), the metric becomes separable, factorization holds, and Bell's inequality is satisfied. For $\gamma > 0$, entropy geometry filters the joint distribution, favoring outcomes that are resolution-stable under curved resolution geometry.

The correlations arise not from hidden signaling or ontological ambiguity, but from structural selection within a curved configuration space. This matches the entropy-weighted stabilization mechanism introduced in Section 3, and is consistent with the collapse interpretation discussed in Section 3.2.

This is the heart of the TEQ interpretation: Bell violations are not mysteries—they are structural shadows of entropy geometry.

For related schematic models, see also [10, 11].

4 Bell Reframed: From Inequality to Curvature

With TEQ, the terrain changes. We no longer have to choose between determinism and influence at a distance. The correlations seen in Bell-type experiments become indicators of shared entropy structure. The statistical violations of Eq. (2) are not paradoxes, but traces of geometric coupling in the configuration space, where factorization assumptions like Eq. (1) no longer hold.

4.1 No Hidden Variables—But No Observer Needed

TEQ dispenses with hidden variables. There is no λ that encodes more information than the wavefunction. But it also avoids the mystical claim that observers cause outcomes. Instead, it asserts that particles become entangled when their respective configurations lie within a shared entropy basin—a region of resolution geometry with coupled curvature.

This shared curvature, introduced via the joint resolution metric in Eq. (7) and exemplified explicitly in Section 3.4, modulates the evolution of the particles even after spatial separation. Not because of a signal or nonlocal interaction, but because they were filtered through a common geometric structure that imprints long-range resolution coupling.

What we call *correlation* is the observable projection of a deeper resolution constraint. This eliminates the need for nonlocal causation: the appearance of influence is an artifact of projecting a curved configuration space onto a flat observational frame. The shared entropy geometry replaces the need for both retrocausality and observer intervention.

4.2 Reproducing Quantum Statistics

The TEQ framework reproduces the statistical structure of quantum theory without postulating it. In particular, the Born rule emerges as a natural consequence of the entropy-weighted path integral introduced earlier (see Eq. (4)):

$$\mathbb{P}[\phi] \propto e^{-\beta g(\phi, \dot{\phi})}.$$

This expression, derived in [9] and justified structurally in [6], mirrors the role of Boltzmann weights in statistical mechanics: trajectories with high entropic cost are suppressed.

Combined with the effective action $S_{\text{eff}}[\phi]$ defined in Eq. (3), this yields a full path selection mechanism that respects entropy curvature. The geometry of resolution, not wavefunction postulates, governs which paths contribute to the ensemble. This includes interference patterns, Bell-type correlations (as in Section 3.4), and entanglement, all arising as entropic consequences of resolution-stable dynamics.

Summary

Bell’s inequality forced a binary choice only within a flat configuration space. TEQ shows that this dichotomy dissolves when the geometry of distinguishability is allowed to curve. The empirical violation of factorization, analyzed in Section 3.3, is not a challenge to realism or locality, but to the assumption that resolution is absolute and globally separable (see Eq. (1)).

Instead, TEQ frames all quantum correlations as surface-level consequences of a deeper entropy geometry. What fails is not realism or causality—but the mistaken presumption of flatness.

5 Implications and Conclusions

Bell’s inequality was never just about numbers—it was about the limits of certain assumptions. TEQ respects the empirical data but shifts the foundation: instead of flat spaces and dichotomies, it offers curvature and constraint.

5.1 What Changes—and What Does Not

All empirical predictions are retained. But the framework beneath them is overhauled. Collapse becomes a statement of local entropy equilibrium: a resolution fixpoint selected not by observation, but by structural suppression of unstable alternatives. Locality is preserved—not in its naive form, but as an emergent property of stable resolution. Realism is not denied, but redefined: what is real is what endures under entropy flow.

What appears as a binary choice between nonlocality and hidden variables dissolves when one questions the geometric assumption of flat resolution. As shown in Section 3 and developed in [6, 1], paths become less likely not because of external causes or observer effects, but because entropy geometry penalizes unstable distinctions. The empirical violation of Eq. (2) thus points to the breakdown of Eq. (1), not because of metaphysical paradox, but because the geometry of distinguishability has curvature (see Section 3.4).

5.2 A Path Beyond Mysticism and Mechanism

TEQ avoids two extremes. It does not postulate consciousness as a causal agent. Nor does it cling to deterministic underlayers or hidden variables. Instead, it builds on entropy geometry, which replaces both mechanistic causality and observer-centric intervention with structural selection under resolution constraints.

Collapse, interference, entanglement—all become features of a dynamic structure in which entropy gradients shape what survives. The violation of Bell’s inequality is no longer a threat. It is a signature. Not of a broken rule, but of a curved space.

5.3 Potential Deviations and Future Tests

While TEQ recovers all standard quantum predictions under weak entropy curvature, it also opens the door to deviations in regimes where entropy gradients become extreme. Because Eq. (4) suppresses high-resolution configurations exponentially, TEQ predicts that fine-grained quantum coherence may break down in settings with large entropy flow or gravitational coupling.

Examples include:

- entangled systems in strongly curved spacetimes,

- delayed-choice or interference experiments with significant entropy asymmetry,
- decoherence rates under structured resolution constraints (e.g., near event horizons),
- entropic modulation of Bell-type correlations in systems with engineered resolution geometries.

These scenarios offer falsifiability: TEQ’s predictions diverge subtly from standard quantum mechanics in cases where entropy curvature is not negligible. This structural sensitivity provides an opportunity for empirical discrimination.

5.4 Falsifiability and Empirical Distinguishability

The following table summarizes scenarios in which TEQ predicts deviations from standard quantum mechanics (see also [6, 1, 12, 13, 14]):

Scenario	Predicted Deviation from QM
Strong entropy gradients (e.g., near event horizons)[12]	Suppression of quantum coherence; altered entanglement statistics
Engineered resolution geometry (lab Bell tests with tunable noise/entropy)[14]	Modulated Bell correlations distinct from standard predictions
Macroscopic quantum states (e.g., large molecules in interference)[13]	Onset of classicality at lower mass/size thresholds than in standard decoherence

5.5 Comparison with Other Interpretations

The TEQ framework diverges from major interpretive camps not by contradiction, but by reduction:

- **Bohmian mechanics** posits a deterministic pilot wave and hidden variables [15, 16]; TEQ requires no such ontology but reproduces statistical results via entropy-weighted path selection.
- **Objective collapse theories** (e.g., GRW) introduce stochastic modifications to dynamics [17, 18]; TEQ derives collapse as a structural fixpoint (see Eq. (5)) without modifying the Schrödinger equation.
- **QBism and relational theories** emphasize the role of the observer and interpret the wavefunction as a subjective degree of belief [19, 20]; TEQ preserves empirically consistent probabilities without invoking subjectivity.

In this way, TEQ provides a unifying perspective: the statistical predictions of quantum mechanics are not postulated, but derived from resolution geometry. The Born rule (Eq. (4)), interference, and entanglement all emerge from entropy-weighted filtering of paths [9].

5.6 Structure Replaces Mystery

Ultimately, TEQ reframes the question. Instead of asking what “causes” collapse, or what lies “behind” entanglement, it asks what kind of structure can preserve resolution in the face of entropy flow. The answer is geometric: curvature governs survival.

The mystery of Bell’s theorem dissolves not by denying experiment, but by upgrading assumption. What fails is not locality, but flatness.

Postlude: On Structure and Surprise

This paper has not introduced new mysteries. It has dissolved one.

The empirical violations of Bell’s inequality once posed a stark metaphysical challenge: either nature allows influence faster than light, or measurement creates reality. But this apparent choice rested on a largely unexamined assumption—that the space of alternatives is flat: that all distinctions are equally real, uniformly separable, and that factorizability holds without structural constraint.

As discussed in the historical background (Section 2), the assumption of globally valid distinguishability was embedded early in quantum formalism but rarely interrogated. TEQ reframes this assumption not as a backdrop, but as a variable geometry shaped by entropy flow.

It shows that distinguishability is not primitive, but selected. That correlation need not arise from causation, but from shared resolution geometry. And that collapse is not a rupture in the fabric of being, but the stabilizing of a fixpoint under entropy flow.

This shift is not interpretive. It is structural: it arises from the internal logic of entropy geometry and resolution filtering, rather than being inserted as an external assumption. By replacing assumptions with derivations, TEQ offers a framework in which known quantum phenomena are recovered as the residue of resolution geometry. But perhaps more importantly, it shows how certain confusions—about the observer, the wavefunction, and nonlocality—arise only when entropy curvature is ignored.

If science is to remain grounded without becoming rigid, and rigorous without denying its own conditions, then geometry must take the place once held by metaphysics. Not as a gesture of humility, but as a statement of structure: what endures, endures because it can be resolved.

The implications reach beyond quantum theory. They concern what can be known, what can be distinguished, and what counts as structure in a world shaped by entropy.

Acknowledgment

This work was carried out independently during a period of cognitive and physical rehabilitation following a brain hemorrhage. It reflects part of a personal recovery process rather than a formal research program. ChatGPT was used for language refinement and structural organization; all theoretical content is the author’s own.

References

- [1] Sigtermans, D. (2025). *The Geometry of Resolution: On Collapse and Structural Stability in Entropy-Based Dynamics*. Preprints.org.
- [2] Hermann, G. (1935). Die naturphilosophischen Grundlagen der Quantenmechanik. *Abhandlungen der Fries'schen Schule*, 6(2).
- [3] Kassner, K. (2017). Grete Hermann's (recovered) solution of the Einstein-Podolsky-Rosen paradox. *European Physical Journal H*, 42, 493–500.
- [4] von Neumann, J. (1932). *Mathematische Grundlagen der Quantenmechanik*. Springer.
- [5] Wigner, E. P. (1959). *Group Theory and its Application to the Quantum Mechanics of Atomic Spectra*. Academic Press.
- [6] Sigtermans, D. (2025). *Entropy as First Principle: Deriving Quantum and Gravitational Structure from Thermodynamic Geometry*. Preprints.org.
- [7] Aspect, A., Dalibard, J., & Roger, G. (1982). Experimental test of Bell's inequalities. *Physical Review Letters*, 49(25), 1804.
- [8] Bell, J. S. (1964). On the Einstein–Podolsky–Rosen paradox. CERN Report.
- [9] Sigtermans, D. (2025). *Deriving Hilbert Space Structure from Entropy Geometry: A Two-Axiom Approach*. Preprints.org.
- [10] Zurek, W. H. (2003). Decoherence, einselection, and the quantum origins of the classical. *Reviews of Modern Physics*, 75(3), 715.
- [11] Brukner, Č., & Zeilinger, A. (2002). Young's experiment and the finiteness of information. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 360(1794), 1061–1069.
- [12] Penrose, R. (1996). On gravity's role in quantum state reduction. *General Relativity and Gravitation*, 28(5), 581–600.
- [13] Arndt, M., & Hornberger, K. (2014). Testing the limits of quantum mechanical superpositions. *Nature Physics*, 10(4), 271–277.
- [14] Giustina, M., et al. (2015). Significant-loophole-free test of Bell's theorem with entangled photons. *Physical Review Letters*, 115(25), 250401.
- [15] Bohm, D. (1952). A suggested interpretation of the quantum theory in terms of “hidden” variables I. *Physical Review*, 85(2), 166–179.
- [16] Bohm, D. (1952). A suggested interpretation of the quantum theory in terms of “hidden” variables II. *Physical Review*, 85(2), 180–193.

- [17] Ghirardi, G. C., Rimini, A., & Weber, T. (1986). Unified dynamics for microscopic and macroscopic systems. *Physical Review D*, 34(2), 470–491.
- [18] Bassi, A., & Ghirardi, G. C. (2003). Dynamical reduction models. *Physics Reports*, 379(5), 257–426.
- [19] Fuchs, C. A., & Schack, R. (2013). Quantum-Bayesian coherence. *Reviews of Modern Physics*, 85(4), 1693–1715.
- [20] Rovelli, C. (1996). Relational quantum mechanics. *International Journal of Theoretical Physics*, 35(8), 1637–1678.